DE–6379 **11**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2013.

CLASSICAL AND STATISTICAL MECHANICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) State Hamilton's variational principle. (5)

   (b) Derive Lagrange's equation for a non-conservative system from variational principle. (10)

   (c) State D'Alembert's principle. (5)

2. (a) Derive Hamilton's equations of motion. (10)

   (b) Write down the canonical equations in terms of the notation of the Poisson brackets. (10)

3. (a) Explain the idea of action and angle variables. (10)

   (b) Write the condition for the transformation to be canonical. (5)

   (c) Derive Hamilton-Jacobi method in canonical transformations. (5)

4. (a) Give the theory of symmetric top and derive the expression for the processional angular velocity. (10)

   (b) Give the principle of small oscillations. (5)

   (c) State the characteristics of a rigid body. (5)

5. (a) Obtain the normal modes of small oscillations. (5)

(b) Describe 'Notational motion'. (5)

(c) Discuss in detail the vibrations of linear triatomic molecule. (10)

6. (a) Show that the Poisson brackets are invariant under canonical transformation. (10)

(b) Write short notes on bound states and parity. (10)

7. (a) Describe Maxwell Boltzmann distribution law for velocities. (10)

(b) Deduce the mean value of energy using law of equipartition of energy. (10)

8. (a) State and prove Liouville's theorem. (10)

(b) Narrate the importance of phase space of statistical systems (5)

(c) Derive the energy fluctuations in canonical ensemble. (5)

DE–6380 **12**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2013.

MATHEMATICAL PHYSICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

Each questions carries 20 marks.

1. (a) Explain the properties of linear vector space. (4)

(b) Find the eigen values and corresponding eigen vectors for the following matrix $\begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}$. (8)

(c) State and prove Cayley-Hamilton theorem. (8)

2. (a) Prove that the gamma function Γz satisfies the relation $\Gamma z \left(\sqrt{1-z}\right) = \frac{\pi}{\sin \pi z}$. (4)

(b) Obtain the series solution of Hermite differential equation $\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2xy = 0$. (8)

(c) Derive the generating function for Legendre polynomial. (8)

3. (a) Show that $\Gamma(n+1) = n\Gamma(n)$ and $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$. (5)

(b) Find the relation between Beta and Gamma function. (5)

(c) Write Rudrigue's formula for legendre polynomial hence find the values of $p_1(x)$ and $p_2(x)$. (10)

4. (a) Find the finite cosine transform of $f(x)$ if (5)

$$f(x) = \frac{\pi}{3} - x + \frac{x^2}{2\pi}.$$

(b) Define Laplace transform. Explain the following (5)

(i) Linearity property

(ii) First shift property.

(c) State and prove Taylor's series theorem. (10)

5. (a) Find out the Laplace transform of $\frac{\sin at}{a}$. (4)

(b) Obtain the polar form of Cauchy-Riemann equations. (8)

(c) Find the solution of Bessel's function

$$x^2\left(\frac{d^2y}{dx^2}\right) + x\left(\frac{dy}{dx}\right) + \left(x^2 - n^2\right) y = 0.$$ (8)

6. (a) Derive heat conduction equation and solve it in two dimensional cartesion coordinates. (4)

(b) Solve the wave equation $\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2}$ for a stretched string fixed by rigid supports at the ends $x = 0$ and $x = l$ if and $t = 0$, $\frac{\partial y}{\partial t} = 0$ and $y(x,t) = y_0(x)$. (8)

(c) Applying the method of variable separation to solve Laplace equation, in cartesian co-ordinates. (8)

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

7. (a) Show that vectors $(u+v)$ $(u-v)$ and $(u-2v+w)$ are linearly independent provided (u, v, w) are linearly independent. (10)

(b) Obtain the expression for orthogonal and unitary transformation. (5)

(c) Prove that orthogonality relation $\int_{-1}^{-1} P_n(x)\, P_m(x)\, dx = 0$ if $m \neq n$. (5)

8. (a) State and prove quotient law of tensors. Show its use with suitable examples. (7)

(b) Show that $\frac{\partial AP}{\partial x^\alpha}$ is not a tensor even through A_p is covariant tensor of rank one. (8)

(c) Show that there is only one group of order 5 and give its multiplication table. (5)

DE–6381 **13**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2013.

INTEGRATED AND DIGITAL ELECTRONICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) Explain emitter follower. (5)

   (b) Define α and β of a transistor, establish the relation between them. (5)

   (c) What is the need for biasing a transistor? Explain (i) fixed bias (ii) voltage divider bias method with neat circuit diagrams. (10)

2. (a) Explain the detail about any three measurements of an operational amplifier parameters. (10)

   (b) Draw the circuit of a summing amplifier and derive the expression for an output voltage. (5)

   (c) Calculate CMRR of an OP-AMP, which has the differential gain of 200000 and the common mode of 6.33. (5)

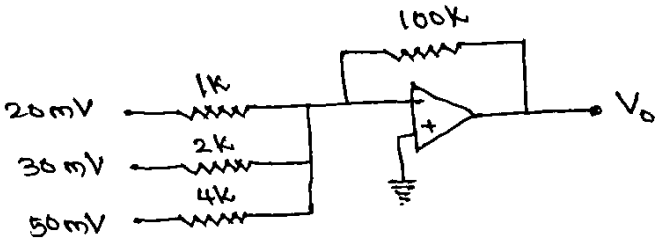
3. (a) Define the following electric parameters.

   (i) CMRR

   (ii) Slew rate

   (iii) Input offset voltage. (6)

(b) Find the output voltage in the following circuit with OP-Amp as summer. (14)



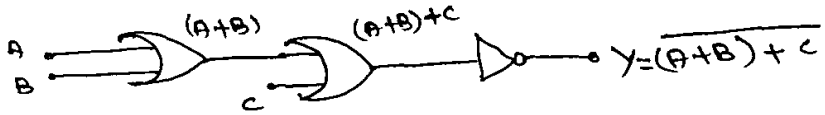
4. (a) Simply the following expressions and then draw the corresponding logic circuits.

(i) $Y = \lfloor A\overline{B}(C + BD) + \overline{AB} \rfloor C$.

(ii) $Y = A\overline{B}D + A\overline{BD}$. (10)

(b) Briefly explain the J-K master slave flip-flop with neat diagram and truth table. (10)

5. (a) Construct the truth table for the following circuit. (10)



(b) Simply the following Boolean expression and draw the simplified logic circuit. (10)

$Y = A\overline{B}C + ABC$.

6. (a) Discuss various types of addressing modes of Intel 8085 with suitable example. (8)

(b) Classify 8085 instructions in various groups. Give examples of instructions for each group. (4)

(c) Explain what is subroutine? What instruction is used to call a subroutine? (8)

7. (a) Draw and explain the timing diagram for (i) fetch (ii) memory read (iii) memory write (iv) I/O read (v) I/O write operations. (10)

(b) Discuss the function of ALU of 8085. (10)

8. (a) What is mean by hall effect? Explain the resistivity measurement using hall effect. (10)

(b) Briefly explain the solar cell fabrication technology with neat diagram and applications. (10)

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DE–6382 **14**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2013.

ELECTRO MAGNETIC THEORY

(2008 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

Each question carries 20 marks.

1. (a) Derive the Maxwell's equations curl $E = \frac{\delta B}{\delta t}$ and $Curl\,\overline{H} = \overline{J} + \frac{\delta \overline{D}}{\delta t}$ respectively. Show that $div\overline{B} = 0$ and $div\overline{E} = V$. (5)

(b) Derive poynting theorem for the conservation of energy in an electro magnetic field and interpret its various terms. (10)

(c) Explain polarization of electromagnetic waves. (5)

2. (a) Discuss the propagation of electromagnetic waves in a isotropic dielectric medium. Show that the waves are of transverse nature. (10)

(b) State and prove poynting theorem. (5)

(c) Derive the Electromagnetic wave equation. (5)

3. (a) How the polarization of electromagnetic wave is affected when it crosses the plane interface between two dielectrics? (8)

   (b) Derive Fresnels equation for reflection and refraction of electromagnetic waves at a plane boundary separating two dielectric media. (8)

   (c) Discuss the phenomenon of total internal reflection. (4)

4. (a) Deduce Brewsters law of the basis of electromagnetic theory and how it can be verified experimentally. (7)

   (b) Explain degree of polarization. (5)

   (c) Discuss the reflection and transmission of electromagnetic waves at the interface of two non conducting media. (8)

5. (a) Derive the scattering cross-section for the scattering of a plane monochromatic electromagnetic radiation by a tree charged particle. (8)

   (b) Deduce clausius Mossotti relation. Point out its limitations. (4)

   (c) What do you mean by dispersion? Discuss it in the case of gaseous medium having both real and complex refractive index. (8)

6. (a) Derive Lorentz-Lorentz formula. (8)

   (b) What is polarization of scattering light? Also discuss about the coherence and incoherence property of scattered light. (8)

   (c) Explain about the dispersion in solids and liquids. (4)

7. (a) Discuss on generation of microwaves. (5)

(b) What is a resonant cavity? Discuss its superiority over resonant circuit for TE waves in the cavity with perfectly conducting walls. (10)

(c) What do you mean by wave guide? (5)

8. (a) What are the conditions for the existence of a plasma? (5)

(b) Discuss the motion of a charged particle in

(i) Uniform electric field.

(ii) Uniform magnetic field. (10)

(c) Explain plasma waves. (5)

DE–6383 **15**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2013.

NUMERICAL METHODS AND PROGRAMMING

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) Give the general format of a structure declaration in 'C'. Give an example. (5)

   (b) What is the purpose of if...else statement? (5)

   (c) Explain about the arithmetic operator in detail. (10)

2. (a) What is meant by constant? Explain it. (5)

   (b) Bring out the differences between while and do-while loops in 'C'. (5)

   (c) Write a program in 'C' to add four matrices. (10)

3. (a) What are advantages of 'C' function? (5)

   (b) How the pointers are used to access the array elements? (5)

   (c) Explain the file creation, writing into and reading from file using fputc( ) and fgetc( ) functions. (10)

4. (a) Explain input/output operations on files. (10)

   (b) Write a 'C' program to multiply on $m \times p$ and a $p \times n$ matrix. (10)

5. (a) Write a 'C' program to find the root of an equation by false position method. (10)

(b) Write a program in 'C' for finding the roots of a equation of the form $x^3 - 3x^2 - 5 = 0$ by Newton-Raphson method. (10)

6. (a) Solve the system by Gauss-Elimination method. (5)

$$2x + 3y - z = 5$$
$$4x + 4y - 3z = 3$$
$$2x - 3y + 2z = 2.$$

(b) Describe Gauss-Seidal method. (5)

(c) The following data give the percentage of Nobel prizes for different age groups. (10)

Age (less than x) :	25	30	40	50
Percentage of Nobel prizes :	52.0	67.3	84.1	94.4

Using Lagrange's formula, estimate the percentage of Nobel prizes under the age of 35.

7. (a) Fit a second degree parabola to the following data : (10)

X :	1	2	3	4	5	6	7	8	9
Y :	2	6	7	8	10	11	11	10	9

(b) Fit a power curve of the form $y = ax^b$ to the following data : (10)

X :	61.0	26.0	07.0	02.6
Y :	350	400	500	600

8. (a) Dividing the range into 10 equal parts find the approximate value of $\int_0^{\pi} \sin x dx$ by

(i) Trapezoidal rule

(ii) Simpson's rule. (10)

(b) By Euler's method solve $\frac{dy}{dx} = y - \frac{2x}{y}$. Initial condition $x_0 = 0$, $y_0 = 1$ in the range $0 \le x \le 0.5$. Take $h = 0.1$. (10)

DE–6384 **21**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2013.

SPECTROSCOPY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Explain the concept of exchange degeneracy. (5)

   (b) What is hybridization ? Explain the formation of sp^2 hybrid orbitals from component orbitals with the figure. (5)

   (c) Outline the Heilter-London wave functions for hydrogen molecule. What are singlet and triplet states of hydrogen. (10)

2. (a) Derive the expression for dissociation energy for a stable diatomic molecule. (10)

   (b) The values of $\bar{v}_e$ and x_e for lower and upper states of C_2 are 1641.4cm^{-1}, 0.00711 cm^{-1} and 1788.2cm^{-1},0.00919cm^{-1}. Find the number of vibrational levels below the dissociation limit and hence the dissociation energy for the lower and upper states of C_2. (1cm^{-1} =11.958J mol^{-1}) (6)

   (c) State Frank-Condon principle. (4)

3. (a) Explain the Raman scattering by

(i) Classical theory and

(ii) Quantum theory. (10)

(b) Rotational and Centrifugal constants of HCl molecule are 10.593 cm^{-1} and 5.3 x 10^{-4} cm^{-1} respectively. Estimate the vibrational frequency and force constant of the molecule. (5)

(c) IR and Raman measurements complement each other and the complete picture of the vibrational problem can only be obtained by using both the techniques. Substantiate. (5)

4. (a) Explain the effect of anharmonicity on the vibrational spectra of diatomic molecule. (10)

(b) What are hot bands? Why are they called so? (6)

(c) The first line in the rotation spectrum of Carbon monoxide has a frequency of 3.8424 cm^{-1} . Calculate the rotational constant and hence bond length in Carbon monoxide. (4)

5. (a) Explain rotational structure of a stokes Raman line of diatomic molecule having a vibrational frequency $\bar{v}_e(1 - 2x_e)$ and illustrate the spectrum. (10)

(b) Write down the characteistics of the hyper Raman scattering selection rules. (5)

(c) Draw the energy level schemes of hyper Rayleigh and hyper Raman scattering. (5)

6. (a) Derive Bloch equations in NMR and transform them into magnetizations and thereby get the expressions for susceptibilities. (14)

(b) Explain ESR spectrum of Hydrogen atom. (6)

7. (a) Draw and explain the energy level schemes and transitions for a MÖssbauer nucleus having $I_g = 1/2, I_e = 3/2$ taking quardrupole interaction into account. (15)

(b) Work out the energy level schemes and resultant quadrupole splitting $I_g = 3/2 \rightarrow I_e = 5/2$ for a system having an axially symmetric electric field gradient tensor. (5)

8. (a) Explain the principle of eletron spin resonance and Why microwave source and techniques have to be applied for the observation of ESR. (5)

(b) Explain ESR spectrum of Hydrogen atom. (10)

(c) Explain the factors responsible for the hyperfine structure of ESR spectra and a free electron is placed in a magnetic field of strength 1.3T. Calculate the resonance frequency if $g = 2.0023$. (5)

DE–6385 **22**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2013.

QUANTUM MECHANICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) State and explain Heisenberg's uncertainity principle. (10)

   (b) Deduce the time-independent Schrodinger's equation from time dependent one. (7)

   (c) Give the physical significance of the wave function. (3)

2. (a) Define probability density. (5)

   (b) Explain in detail, the rigid rotator motion and solve the Schrodinger equation of it. (10)

   (c) Define parity. Give examples. (5)

3. (a) Define a Hermitian operator and state any three properties of it. (5)

   (b) Show that WKB approximation gives the connection formulae and thus the quantization condition. (10)

   (c) Calculate the energy of the ground state of Helium atom. (5)

4. (a) Establish Schrodinger equation for a linear harmonic oscillator and solve for its eigen values and eigen functions. (10)

   (b) Discuss the significance of zero point energy. (5)

   (c) Explain Schwartz inequality. (5)

5. (a) Give the first order perturbation theory for a non-degenerate case. (10)

   (b) Discuss in detail "Stark effect" in the context of perturbation theory. (10)

6. (a) Develop the semi-classical radiation theory and use it to determine the conditions for allowed transitions. (10)

   (b) Write short notes on parity operator. (5)

   (c) Write short notes on unitary operator. (5)

7. (a) Describe how scattering amplitude and scattering cross-section can be obtained using born approximation. (10)

   (b) Write down the salient features of optical theorem. (5)

   (c) Give the details of stimulated emission. (5)

8. (a) Obtain the commutation relation for L_x, L_y and L_z, the components of angular momentum operator, show that L^2 commute with any of the three components. (10)

   (b) Define C.G. co-efficients. Discuss the construction procedure for the C.G. co-efficient matrix. (10)

DE–6386 **23**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2013

SOLID STATE PHYSICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carries 20 marks.

($5 \times 20 = 100$)

1. (a) Give the basic classification of crystal systems. (10)

   (b) Explain the Farier analysis of the basis. (10)

2. (a) Discuss about

   (i) Ionic crystals and

   (ii) Covalent crystals. (10)

   (b) Sketch the elastic waves in cubic crystals with necessary theory. (10)

3. (a) Derive expression for dispersion relation for vibrations with two atoms per primitive basis. (10)

   (b) Describe the inelastic scattering of phonons. (10)

4. (a) Obtain expression for heat capacity of the electron gas. (10)

   (b) Solve the wave equation of electron in a periodic potential. (10)

5. Write a note on

   (a) Semimetals. (8)

   (b) Super lattices. (7)

   (c) Nano structures. (5)

6. Write an essay on Fermi surfaces and metals. (20)

7. (a) Derive Langerain in equation for diamagnets. (10)

   (b) Discuss the quantum theory of paramagnetism. (10)

8. Explain

   (a) Ferromagnetic domains. (7)

   (b) Magnetic bubble domains. (7)

   (c) Ferrimagnetic order. (6)

DE–6387 **24**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2013.

NUCLEAR AND PARTICLE PHYSICS

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Describe the Fermi's theory of $\beta-$ decay. (8)

   (b) Explain the Nuclear Isomerism. (8)

   (c) Write a note on Election capture (4)

2. (a) Discuss the experimental method of the parity violation in $\beta-$decay. (8)

   (b) Briefly explain about the Neutrinos. (4)

   (c) Discuss the Gamnia decay. (8)

3. (a) What are Magic numbers? Give its significance. (6)

   (b) Deduce Bohr –Wheeler's theory of nuclear fission. (10)

   (c) Write a note on Collective model. (4)

4. (a) Explain the theory of Nuclear shell model and indicate the predictions involved in the shell model. (14)

   (b) Briefly explain the Electric quadrapole moment. (6)

5. (a) Explain the determination of phase-shift. (6)

(b) What are called Tensor forces? (2)

(c) Discus the Yukawa's meson theory of nuclear forces and also drive the relation for Yukawa's potential. (12)

6. (a) Explain the resonance scattering cross section. (6)

(b) Describe the compound nucleus model with examples. (14)

7. (a) Discuss the various types of Nuclear reactors. (15)

(b) Write a note on thermal Neutrons. (5)

8. (a) Discuss about the strong and weak interactions. (6)

(b) Explain in detail about charge conjugation, parity and time reversal and the combination of them with suitable examples. (14)

DE–6388 **25**

DISTANCE EDUCATION

M.Sc. (Physics) DEGREE EXAMINATION, MAY 2013.

MATERIALS SCIENCE

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

1. (a) Describe the elastic and inelastic behaviour of materials. (15)

   (b) Write a note on polymers. (5)

2. (a) Describe the construction and working of a semi conductor laser with a neat diagram. (14)

   (b) Briefly explain the population rate equation. (6)

3. (a) Discuss about the optical resonator. (8)

   (b) Explain in detail about the Dye laser. Give its applications. (12)

4. (a) With a suitable example explain the electro-optical effect. (8)'

   (b) What is meant by non-linear optics? How are second harmonic waves generated? (12)

5. (a) Describe the principle, theory and working of diffusion pump. (12)

   (b) Classify the sputtering methods. (8)

6. (a) Explain the principle and working of Penning gauge. (12)'

   (b) Describe the Quartz crystal method. (8)

7. (a) Explain the thickness measurement of thin films. (8)

   (b) What are copper ion conductors? Give its applications. (12)

8. (a) Describe the production and measurement of high vacuum. (8)

   (b) Explain in detail the electrochromic display systems. (12)